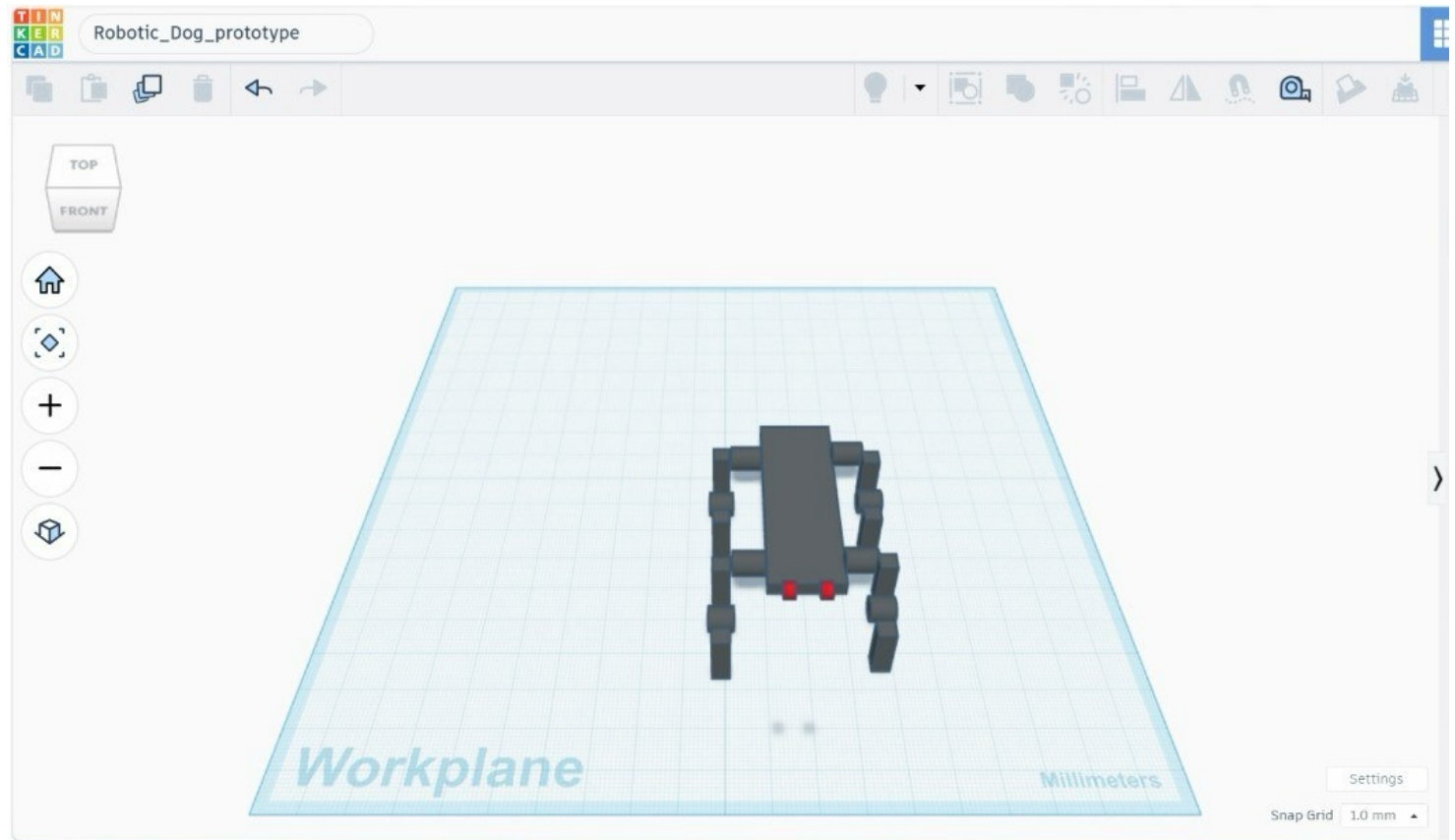


Robotic Dog Design Report



- **Design Concept:** The robot has a simple, compact body with four legs. The design is lightweight and balanced to improve stability during movement. The body also provides enough space for the battery and electronic components.
- **Motors:** I chose **Servo Motors** because they are small, accurate, and suitable for controlling the leg joints in a prototype robot.
- **Joints and Degrees of Freedom (DOF):** Each leg has **2 Degrees of Freedom (DOF)**: one joint at the hip and one at the knee. Since the robot has four legs, the total is **8 DOF**, which is enough for basic walking.
- **Stability:** The robot's center of gravity is located near the middle of the body. This helps keep the robot balanced and reduces the chance of tipping over while walking.
- **Movement (Gait):** The robot uses a **Crawl Gait**, where one leg moves at a time while the other three remain on the ground. This provides good stability during movement.
- **Torque Calculation:** To estimate the required torque for one leg joint, I used the following equation:

$$\tau = F \times d$$

Where:

- **τ (Torque):** Required torque (N·m)
- **F (Force):** Weight of the leg $\approx 1 \text{ N}$
- **d (Distance):** Length of the leg = **0.05 m**

Calculation:

$$1 \text{ N} \times 0.05 \text{ m} = 0.05 \text{ N}\cdot\text{m}$$

- **Challenges:** The main mechanical challenge is slipping on smooth surfaces, so rubber pads can be added to the feet for better grip. Another challenge is keeping the battery weight balanced so the robot remains stable while walking.